

संभव BATCH

Answer Key & Solutions: Geography Hard Module

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Zero Error Verified | Detailed Analytical Explanations

Question 1

Correct Answer: Both A and R are true and R is the correct explanation of A.

Detailed Explanation:

The Jovian (gas giant) planets retained their thick, primordial atmospheres of Hydrogen and Helium because the solar wind was not intense enough at that distance to strip them away, unlike the terrestrial planets.

Question 2

Correct Answer: 1 and 3 only

Detailed Explanation:

Statement 2 is incorrect because according to Kepler's laws of planetary motion, the orbital velocity of planets DECREASES as their distance from the Sun increases (Mercury is the fastest, Neptune is the slowest).

Question 3

Correct Answer: A-3, B-2, C-4, D-1

Detailed Explanation:

Mercury lacks an atmosphere (3). Venus is the hottest due to a runaway greenhouse effect (2). Jupiter has the Great Red Spot (4). Saturn has the lowest density, less than water (1).

Question 4

Correct Answer: It is heavily dependent on the intense solar flares and coronal mass ejections of the star.

Detailed Explanation:

The 'Goldilocks Zone' refers to the habitable zone around a star where conditions are just right for liquid water. Intense solar flares and coronal mass ejections are extremely detrimental to habitability and atmosphere retention; they do not define the zone.

Question 5

Correct Answer: 2, 4, 1, 3

Detailed Explanation:

The correct chronological sequence after the Big Bang is: Inflation/rapid expansion (2), Nucleosynthesis forming the first nuclei (4), Recombination forming neutral atoms (1), and finally the formation of the first stars and galaxies (3).

Question 6

Correct Answer: Both A and R are true and R is the correct explanation of A.

Detailed Explanation:

A comet's glowing tail is a direct result of its proximity to the Sun. Solar radiation vaporizes its volatile ices, and the solar wind pushes this material outward, creating a tail that always points away from the Sun.

Question 7

Correct Answer: A-2, B-3, C-4, D-1

Detailed Explanation:

Ganymede is the largest moon of Jupiter (2). Titan is the largest moon of Saturn (3). Triton is a large moon of Neptune (4). Phobos is a small moon of Mars (1).

Question 8

Correct Answer: 1, 2 and 3

Detailed Explanation:

All statements are correct. The photosphere is the visible surface (approx. 5,500°C), while the outer corona is mysteriously much hotter (millions of degrees). Sunspots are relatively cooler regions caused by intense magnetic flux.

Question 9

Correct Answer: Pluto

Detailed Explanation:

In 2006, the International Astronomical Union (IAU) redefined the term 'planet'. Pluto was reclassified as a 'dwarf planet' because it has not 'cleared its neighboring region of other debris' (the Kuiper belt).

Question 10

Correct Answer: Heliopause

Detailed Explanation:

The Heliopause is the theoretical boundary where the Sun's solar wind is ultimately stopped by the interstellar medium, marking the edge of the heliosphere.

Question 11

Correct Answer: Both A and R are true and R is the correct explanation of A.

Detailed Explanation:

Igneous rocks are termed primary rocks because they are the first rocks to form directly from the cooling and solidification of molten magma or lava, from which all other rocks are subsequently derived.

Question 12

Correct Answer: A-2, B-1, C-4, D-3

Detailed Explanation:

Granite is an intrusive igneous rock (A-2). Basalt is an extrusive igneous rock (B-1). Sandstone is a clastic sedimentary rock (C-4). Gneiss is a foliated metamorphic rock (D-3).

Question 13

Correct Answer: 3 only

Detailed Explanation:

Statement 3 is NOT correct. Limestone and chalk are predominantly chemically or organically formed sedimentary rocks, not clastic (which are formed from mechanical weathering debris like sandstone and shale).

Question 14

Correct Answer: Olivine

Detailed Explanation:

According to Bowen's Reaction Series, Olivine is the first ferromagnesian silicate mineral to crystallize from a mafic melt at extremely high temperatures, followed by pyroxene, amphibole, and biotite.

Question 15

Correct Answer: A-2, B-3, C-1, D-4

Detailed Explanation:

Under heat and pressure, Limestone metamorphoses into Marble (2), Sandstone into Quartzite (3), Shale into Slate (1), and Granite into Gneiss (4).

Question 16

Correct Answer: Both A and R are true and R is the correct explanation of A.

Detailed Explanation:

Basaltic (mafic) lava has a low silica content and erupts at very high temperatures, which gives it a very low viscosity. This allows it to flow easily over long distances, forming shield volcanoes or lava plateaus.

Question 17

Correct Answer: Coal

Detailed Explanation:

Coal is a classic example of an organically formed sedimentary rock, created from the accumulation and lithification of plant debris in swampy environments over millions of years.

Question 18

Correct Answer: Both 1 and 2

Detailed Explanation:

Extrusive rocks cool quickly on the surface, forming small, fine crystals (aphanitic). Intrusive rocks cool slowly deep underground, allowing large, visible crystals to form (phaneritic).

Question 19

Correct Answer: Heat and Pressure directly without weathering

Detailed Explanation:

An igneous rock can bypass the sedimentary stage and be converted directly into a metamorphic rock if it is subjected to intense heat and pressure deep within the Earth's crust.

Question 20

Correct Answer: Metamorphic rocks

Detailed Explanation:

Foliation is a textural feature exclusive to metamorphic rocks, where directed pressure causes minerals to align themselves in parallel, banded, or sheet-like layers (e.g., schist, gneiss).

Question 21

Correct Answer: A is true but R is false.

Detailed Explanation:

Assertion (A) is true (Shield volcanoes like Mauna Loa are huge but effusive). Reason (R) is false because they are constructed of low-viscosity, BASALTIC lava (low silica), not highly viscous silicic lava.

Question 22

Correct Answer: A-3, B-4, C-1, D-2

Detailed Explanation:

Batholith is a massive deep-seated magma body (3). Laccolith is a mushroom-shaped intrusion that blisters the overlying rock (4). Dyke is a vertical intrusion (1). Sill is a horizontal intrusion (2).

Question 23

Correct Answer: They are formed by the slow, continuous effusion of low-viscosity lava.

Detailed Explanation:

This statement is NOT correct. Calderas are vast, depression-like craters formed by highly explosive eruptions that rapidly empty the magma chamber, causing the volcano's summit to collapse inward.

Question 24

Correct Answer: 2 only

Detailed Explanation:

Composite volcanoes, also known as stratovolcanoes (e.g., Mt. Fuji, Mt. St. Helens), are characterized by a steep profile built up by alternating layers (strata) of hardened lava, tephra, pumice, and volcanic ash.

Question 25

Correct Answer: Hotspot (Mantle plume) volcanism

Detailed Explanation:

The Hawaiian Islands are located far from any plate boundaries. They are formed by a stationary 'hotspot' or mantle plume, over which the Pacific Plate moves, creating a chain of volcanic islands.

Question 26

Correct Answer: A is true but R is false.

Detailed Explanation:

Assertion (A) is true. Reason (R) is false because the Pacific 'Ring of Fire' is a continuous series of CONVERGENT plate boundaries (subduction zones), not divergent boundaries.

Question 27

Correct Answer: Water vapour

Detailed Explanation:

Water vapour (H₂O) is by far the most abundant volcanic gas emitted during an eruption, typically making up 50% to 90% of the total gas volume, followed by Carbon dioxide and Sulphur dioxide.

Question 28

Correct Answer: Pelean eruption

Detailed Explanation:

A Pelean eruption is characterized by highly viscous, gas-rich felsic magma that frequently produces devastating pyroclastic flows (glowing avalanches of hot ash and gas).

Question 29

Correct Answer: A-2, B-3, C-1

Detailed Explanation:

Volcanic Ash consists of very fine particles less than 2 mm (2). Lapilli are pebble-sized fragments between 2 mm and 64 mm (3). Volcanic Bombs are large masses greater than 64 mm (1).

Question 30

Correct Answer: Phacolith

Detailed Explanation:

A Phacolith is a lens-shaped or saucer-shaped intrusive igneous rock body that occupies the crest of an anticline or the bottom of a syncline in folded sedimentary strata.

Question 31

Correct Answer: 1 and 2 only

Detailed Explanation:

Statement 3 is incorrect. Weathering and erosion are classic examples of EXOGENIC (external) processes driven by solar energy and gravity, not endogenic processes.

Question 32

Correct Answer: Both A and R are true and R is the correct explanation of A.

Detailed Explanation:

Chemical weathering processes heavily rely on the presence of water and high temperatures to drive the chemical reactions. Therefore, hot and humid (tropical) climates experience maximum chemical weathering.

Question 33

Correct Answer: Carbonation

Detailed Explanation:

Carbonation is a form of CHEMICAL weathering where carbon dioxide dissolves in water to form weak carbonic acid, which then dissolves susceptible rocks like limestone. The others are physical processes.

Question 34

Correct Answer: A-2, B-1, C-4, D-3

Detailed Explanation:

Creep is extremely slow soil movement (2). Solifluction occurs over permafrost (1). Mudflow is rapid fluid flow in channels (4). Rockfall is free-falling bedrock (3).

Question 35

Correct Answer: Unloading (Exfoliation)

Detailed Explanation:

Unloading, or exfoliation, is a physical weathering process where the removal of overlying rock reduces pressure, causing the underlying rocks (like granite) to expand and fracture into onion-like sheets.

Question 36

Correct Answer: 2 only

Detailed Explanation:

Statement 2 is NOT correct. Weathering is a STATIC (in-situ) process that involves the breakdown of rocks without the active transportation of the weathered materials. Transportation is the defining feature of erosion.

Question 37

Correct Answer: Physical weathering due to temperature changes and wind.

Detailed Explanation:

In arid (desert) regions, there is little to no water for chemical weathering (like carbonation). Therefore, physical/mechanical weathering (due to extreme diurnal temperature changes and wind abrasion) dominates.

Question 38

Correct Answer: A is false but R is true.

Detailed Explanation:

Assertion (A) is false because Solifluction occurs predominantly in Tundra or periglacial climates, not tropical rainforests. Reason (R) correctly explains that it requires a permafrost layer.

Question 39

Correct Answer: Dense vegetation cover on slopes

Detailed Explanation:

Dense vegetation acts as a stabilizing agent. The roots bind the soil together and absorb excess moisture, thereby PREVENTING or significantly reducing the likelihood of mass wasting.

Question 40

Correct Answer: A-2, B-3, C-1, D-4

Detailed Explanation:

Oxidation involves oxygen reacting with minerals (2). Carbonation involves CO₂ and water (3). Hydration involves water integrating into a mineral's crystalline structure (1). Solution is direct dissolving (4).

Question 41

Correct Answer: Statement I is correct, Statement II is incorrect.

Detailed Explanation:

Diastrophism encompasses all processes that move or deform the solid crust (folding, faulting, orogeny). Volcanism is the movement of molten magma onto or toward the surface and is distinctly categorized separately from diastrophism.

Question 42

Correct Answer: Extremely rapid mass movement of unconsolidated rock, soil, and debris.

Detailed Explanation:

A debris avalanche is a sudden, extremely rapid, and highly destructive type of mass movement (landslide) characterized by the chaotic flow of mixed rock, soil, and debris.

Question 43

Correct Answer: A is false but R is true.

Detailed Explanation:

Assertion (A) is false because in their youth stage, rivers primarily engage in VERTICAL erosion (down-cutting), forming V-shaped valleys, not lateral erosion. Reason (R) is the correct explanation for why down-cutting occurs.

Question 44

Correct Answer: Wind

Detailed Explanation:

Mushroom rocks (or Zeugen/Pedestal rocks) are classic Aeolian (wind) landforms created by the abrasive action of wind-blown sand, which erodes the softer lower portions of a rock mass faster than the harder top.

Question 45

Correct Answer: Accumulations of dirt and rocks that have fallen onto the glacier surface or have been pushed along by the glacier as it moves.

Detailed Explanation:

Moraines are glacial depositional features consisting of unstratified debris (till) left behind by a moving or retreating glacier (e.g., terminal, lateral, or medial moraines).

Question 46

Correct Answer: 2 only

Detailed Explanation:

Statement 2 is NOT correct. While stalactites and stalagmites are depositional features, Sinkholes are strictly EROSIONAL/destructional features formed by the dissolution of limestone bedrock.

Question 47

Correct Answer: A-4, B-2, C-1, D-3

Detailed Explanation:

River action forms V-shaped valleys (A-4). Glaciers form U-shaped valleys (B-2). Wind forms Barchan dunes (C-1). Sea Waves form Sea Stacks (D-3).

Question 48

Correct Answer: Mechanical weathering

Detailed Explanation:

Mechanical (physical) weathering involves the physical disintegration of rocks into smaller fragments (via frost wedging, thermal expansion, etc.) without altering their fundamental chemical composition.

Question 49

Correct Answer: A is false but R is true.

Detailed Explanation:

Assertion (A) is false. Stalactites grow DOWNWARDS from the ceiling of caves (remember 'c' for ceiling). Stalagmites grow upwards from the ground (remember 'g' for ground). Reason (R) accurately describes their chemical formation.

Question 50

Correct Answer: Weathering involves the in-situ breakdown of rocks, while erosion involves the transportation of the broken materials.

Detailed Explanation:

Weathering happens in place (in-situ) without moving the material. Erosion is the subsequent acquisition and transportation of that weathered debris by dynamic agents like water, wind, or ice.